

Leibniz’s law of continuity, read on records, selects a positive action floor for motions

Result, 2026-09-29 (Claude; written derivation and passage readings, unrefereed).

In the mark model of the Planck paper (§3), the least error with which any admissible protocol decides Newton’s comparison, inertia against a constant force F over a cell of duration τ with unknown initial position and velocity, is

$$P_*(F) = \Phi\left(-\frac{1}{2}\sqrt{K_\tau/\kappa}\right), \quad K_\tau = \frac{F^2\tau^3}{24m},$$

for a mark floor $\kappa > 0$, while $P_*(F) = 0$ for every $F \neq 0$ when $\kappa = 0$; at $F = 0$ the cases coincide and $P_* = \frac{1}{2}$. **The best verdict is continuous in the force at $F = 0$ if and only if $\kappa > 0$.** The same holds on Zeno’s rung, rest against uniform motion, with $mv^2\tau/(8\kappa)$ in place of $K_\tau/(4\kappa)$. Static shapes behave differently: there the best verdict jumps at zero for every κ , because reading a shape disturbs nothing.

Leibniz stated the law of continuity in July 1687, the year of the *Principia*, as a test of laws of motion: when two cases approach and are lost in one another, their outcomes must do the same, and he describes the nearly equal case of a collision as one “dont à peine ce cas peut estre distingué” (§2). Read with outcomes measured by how well they can be told apart, the law holds for Newton’s comparison exactly when $\kappa > 0$. Read with outcomes measured by positions and velocities, it holds on both branches. The fork that every route of this programme reaches (positivity as a premise, the zero branch admissible) is therefore located inside one Newton-age text, and the programme’s thesis, that quantization is a consistency condition on laws which Newton’s continuum limit missed, has a period form: Leibniz’s “preuve ou examen” of laws of motion, applied to their records. His *petites perceptions* (1704) supply the graded, additive discernibility that the records reading needs (§3). The Newton-age dispute over indivisibles, Galileo, Cavalieri and Guldin between 1621 and 1647, concerned static figures, where the theorem imposes no floor and Newton’s limit doctrine is the right answer (§4).

1. The proposition

Setting. The statistical model of the Planck paper, §3: mark j at time t_j returns the position with Gaussian error of standard deviation δ_j and delivers an independent Gaussian impulse of standard deviation Δ_j ; every available mark obeys $\delta_j\Delta_j \geq \kappa$; protocols are non-adaptive and tests are invariant under the unknown y_0, v_0 . The optimal invariant test errs with probability $\Phi(-d/2)$, and Theorem 2 of the paper gives $d^2 \leq K_\tau/\kappa$ for every protocol, with a sharp constant.

Proposition L. Let $P_*(F)$ be the infimum of the error over admissible protocols and invariant tests. Then $P_*(F) = \Phi(-\frac{1}{2}\sqrt{K_\tau/\kappa})$ if $\kappa > 0$, and $P_*(F) = 0$ for $F \neq 0$ if $\kappa = 0$. For Zeno’s rung (rest against speed v , unknown initial position), $P_*(v) = \Phi(-\sqrt{mv^2\tau/(8\kappa)})$ if $\kappa > 0$ and $P_*(v) = 0$ for $v \neq 0$ if $\kappa = 0$. For a static taper ϑ read at two heights, $P_*(\vartheta) = 0$ for $\vartheta \neq 0$ and every $\kappa \geq 0$.

Proof. For $\kappa > 0$, $\Phi(-d/2)$ decreases in d , and the supremum of d^2 over protocols is K_τ/κ by the sharpness in Theorem 2; hence the infimum of the error. For $\kappa = 0$, marks with $\Delta_j = 0$ and arbitrary $\delta_j > 0$ are admissible; take three marks at distinct times and weights u with $\sum u_i = \sum u_i t_i = 0$, so that $u^\top P = \frac{F}{2m} \sum u_i t_i^2 \neq 0$ while $u^\top \Sigma u = \sum \delta_i^2 u_i^2 \rightarrow 0$; then $d \rightarrow \infty$. Zeno’s rung is Theorem 9(i) of the paper, whose bound $d^2 \leq mv^2\tau/(2\kappa)$ two marks attain. The static case is Theorem 9(iii): with N marks of resolution δ at each height, $d^2 = N\vartheta^2\Delta h^2/(2\delta^2)$, unbounded in N whatever κ is. $\square$

For $\kappa > 0$ the verdict is Lipschitz at $F = 0$ with constant $\tau^{3/2}/(2\sqrt{2\pi}\sqrt{24m\kappa})$, which diverges as $\kappa \rightarrow 0$: the zero branch is the singular limit of a family of continuous verdict functions, the pattern of an order of limits. Quantum mechanics agrees with the model on the split between the two kinds of case. A static parameter can be estimated without limit by repetition, while

the dynamical comparison in a fixed window has the N -independent bound that the paper’s quantum results supply.

2. Leibniz, 1687: the law of continuity as a test of laws

Leibniz, “Lettre de M. L. sur un principe general utile à l’explication des loix de la nature”, *Nouvelles de la République des Lettres*, July 1687, in Gerhardt, *Die philosophischen Schriften* III (1887), 51–55 (companion; passage, OCR normalized by hand).

- **The text.** “Lorsque la difference de deux cas peut estre diminuée au dessous de toute grandeur donnée *in datis* ou dans ce qui est posé, il faut qu’elle se puisse trouver aussi diminuée au dessous de toute grandeur donnée *in quaesitis* ou dans ce qui en resulte ... *Datis ordinatis etiam quaesita sunt ordinata*” (p. 52). “Le repos peut estre considéré comme une vistesse infiniment petite, ou comme une tardité infinie”, so that the rule of rest must be a particular case of the rule of motion, “autrement ... ce sera une marque assurée, que les regles sont mal concertées” (pp. 52–53). Against Descartes’s second rule of collision, a body B made “aussi petite que l’on voudra” larger than C should reflect a little less, and C a little more, “qu’au cas de l’égalité dont à peine ce cas peut estre distingué” (p. 53). And the exemption: “dans les choses composées quelques fois un petit changement peut faire un grand effect, comme par exemple une estincelle tombant dans une grande masse de la poudre à canon”, while “à l’égard des principes ou choses simples, rien de semblable ne sçauroit arriver” (p. 54).
- **What it commits him to.** A law of motion is tested from outside, before any inner discussion (“preuve ou examen ... avant même que de venir à une discussion interieure”), by whether its outcomes converge when its cases do. Rest is the limit of slow motion and equality the limit of inequality, and a rule that jumps at either is ill made. Composite amplifiers may magnify small differences into large effects; simple principles may not jump.
- **What the theorem does with it.** Leibniz applies the test to collision velocities, the geometric reading, and Newton’s laws pass it on both branches. His own phrase for the nearly equal case, however, is a statement about discernibility, and Proposition L shows what the test demands when outcomes are measured that way: for Newton’s comparison and for Zeno’s rung, continuity holds exactly when $\kappa > 0$. The exemption marks the boundary of the argument. An instrument is a composite amplifier, and finite amplification is allowed; the zero branch needs the supremum over all instruments the laws permit to jump, a jump in what the laws allow to be recorded, which is the level of principle if the mark trade-off is a law. Whether the verdict of the best possible record counts among “ce qui en resulte” is the premise, now stated in Leibniz’s terms. The theorem also declines one extension: for static shapes the records reading fails on both branches, and Leibniz’s letter applies the law to figures geometrically (the ellipse passing into the parabola, p. 52), which Proposition L shows to be the only reading that holds there.

Prior art, at metadata level: Mortensen, “The Leibniz Continuity Condition, Inconsistency and Quantum Dynamics”, *J. Philos. Logic* **26** (1997), 377–389, treats the condition against quantum jumps at the instant of change; the pairing of continuity with the identity of indiscernibles is the subject of Russell’s chapter “The Identity of Indiscernibles and the Law of Continuity” in *A Critical Exposition of the Philosophy of Leibniz*. The operational reading used here is the base-norm topology on states, in which classical pure states are pairwise perfectly distinguishable; compare the Hardy audit, whose continuity axiom concerns reversible paths between pure states.

3. Leibniz, 1704: small perceptions add

Leibniz, *Nouveaux essais*, Préface, written 1703–1705, in Gerhardt V (1882), 47–49 (companion; passage, OCR normalized by hand).

- **The text.** To hear the roar of the sea “il faut bien qu’on entende les parties qui composent ce tout, c’est à dire les bruits de chaque vague ... Car il faut qu’on en soit affecté un peu par le mouvement de cette vague ... autrement on n’auroit pas celle de cent mille vagues, puisque cent mille riens ne sauroient faire quelque chose” (p. 47). “La nature ne fait jamais des sauts ... jamais un mouvement ne naist immédiatement du repos ny s’y réduit que par un mouvement plus petit” (p. 49). “En vertu des variations insensibles, deux choses individuelles ne sauroient estre parfaitement semblables” (p. 49).
- **What it commits him to.** Every effect, however small, registers a little; perception is graded and additive; noticeable perceptions come by degrees from insensible ones; distinct individuals always differ, by differences that can be insensible.
- **What the theorem does with it.** The cut measure makes the addition exact. Newton’s cell action K_τ is spent additively over any sequence of cuts, a share $3s(1-s)K$ per cut at fraction s (cut-measure note, Theorems 1–2), and it is the Cameron–Martin energy of the force’s shift (Proposition 4), so each cut carries a finite share of the whole comparison’s discernibility and the shares add to K_τ/κ . With $\kappa > 0$ each cut is a *petite perception* of the force, insensible alone below the mark mesh and effective in the assembly; with $\kappa = 0$ a single cut decides, and records of motion have no insensible parts. The zero branch can still place insensibility in the perceiver, as Leibniz’s confusion of created substances does, graded from one substance to another. The stochastic route shows how composition turns graded constants into one ($\kappa = mD$ for every body), so the step from “every created perceiver is confused” to one universal floor is available as a reading; it is recorded here as a reading. The maxim on motion from rest is Galileo’s passage through all degrees of slowness, stated at the beginning of motion, the regime of Lemma X (“ipso motus initio”); Zeno’s rung of Proposition L is its records version.

4. The Newton-age dispute over indivisibles, 1621–1647

Book I’s closing scholium prefers limits to indivisibles, “quoniam durior est indivisibilium Hypothesis”, without naming the dispute it settles. The primary texts are held in docs/classics.

- **Galileo’s bowl and cone** (*Discorsi*, Giornata prima, Favaro VIII, 74–76, 78–80; companion). Every horizontal section of the bowl (cylinder minus hemisphere) equals the section of the inscribed cone, so the surfaces are “sempre eguali, e ... diminuendosi sempre egualmente, vadano a terminare l’una in un sol punto e l’altra nella circonferenza d’un cerchio ... perché in questa conseguenza sola versa la nostra maraviglia”. His conclusion: “questi attributi di maggioranza, minorità ed egualità non convenghino a gl’infiniti”. *Commitment*: comparison fails among infinities and indivisibles. *The theorem*: the bowl is Democritus’s third (bowl and cone are each a third of the cylinder), a static figure, so Theorem 9(iii) and Proposition L put no floor on it, and Newton’s sentence that ultimate ratios are limits and never ratios of ultimate quantities dissolves the wonder: the ratio of sections is 1 at every cut, and point and circle are never compared. Leibniz reverses Galileo’s conclusion in 1687: equality is an infinitely small inequality, so the attributes extend by continuity.
- **Cavalieri’s reply** (letter 2992 of 1634, Opere XVI 136–138; companion). Equal parts are removed “essendo noi arrivati al nullo piano tanto nel cono quanto nella scodella”, and “non mi dichiaro di componere il continuo d’indivisibili, ma solo mostro che i continui hanno la proportionione delli aggregati di questi indivisibili”. *Commitment*: ratios of aggregates of sections, with no claim of composition. *The theorem*: the cut measure keeps

exactly this stance, a statement about aggregates of cuts (shares summing to K_τ for every schedule) that composes nothing, and the mark mesh bounds what can be exhibited without bounding what can be cut.

- **Galileo’s wheel** (Favaro VIII, 68–72, same companion). A smaller concentric wheel covers the larger wheel’s line with its own sides “con l’interposizione di cento mila spazii vacui traposti”, and for circles “sì come i lati non son quanti, ma bene infiniti, così gl’interposti vacui non son quanti, ma infiniti”; Simplicio hears in it “quei vacui disseminati di certo filosofo antico”, and Salviati’s retort about the denier of Providence names Epicurus. *Commitment*: a continuum composed of infinitely many unquantified indivisibles, partly full and partly void, crossed by leaps across the voids. *The theorem*: Galileo publishes al-Nazzam’s leap in 1638 as consistent. The paper’s reading of the leap applies: below the mesh a record is consistent with a leap, the interval stays divisible, and the crossing is exhibited in finitely many steps.
- **Guldin** (*Centrobaryca* IV, 1641, preface p. 4; companion): “Galileus profecto in eodem Dialogo de Motu locali, disputans de infinito, de proprietatibus finitorum, quas infinitis applicare minime liceat, contra ipsum concludit.” *Commitment*: Galileo’s own principle refutes Cavalieri. *The theorem*: it sides with Guldin and Newton on composition (no least magnitude follows from κ) and with Cavalieri on ratios of aggregates.

The four entries share one feature that sharpens the paper’s §9. The seventeenth-century dispute Newton answered concerned static figures, and there the theorem agrees with his answer and imposes nothing. The floor enters only where a record carries back-action, in the lemmas applied to motion, and the dispute about motion is the one he passed over.

5. Consequence for STATE

For the Newton goal: positivity of the action floor is equivalent, in the mark model, to Leibniz’s 1687 law of continuity for laws of motion with outcomes measured by discernibility (Proposition L). The zero branch keeps the geometric reading, which Leibniz himself applied, so the fork stands; it is now a choice between two readings of one Newton-age principle, with Leibniz’s phrase “à peine distingué” and his doctrine of small perceptions on the side of the floor, and statics showing why the reading must be confined to motions. For the scholion: two Leibniz entries and the Galileo–Cavalieri–Guldin layer are supplied with their three obligations; a pointer belongs in the paper’s §9. To be refereed (Astra or Fable): the proposition, the passage readings, and whether the records reading is fair to Leibniz.